E-Governance Grievance Redressal & Escalation Workflow System

JAVA PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF **BACHELOR OF ENGINEERING** IN **ARTIFICIAL INTELLIGENCE AND DATA SCIENCE** OF THE ANNA UNIVERSITY

JAVA PROJECT
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PROJECT WORK

Submitted by

SHANMUGAPRIYAN D - 722825243197

BATCH
2025 – 2029

Under the Guidance of

Dr.C.Rajeshkumar

Assistant Professor

Department of Computer & Communication Engineering

Sri Eshwar College of Engineering

(An Autonomous Institution – Affiliated to Anna University)

COIMBATORE – 641 202

Sri Eshwar College of Engineering

(An Autonomous Institution – Affiliated to Anna University)
COIMBATORE – 641 202

BONAFIDE CERTIFICATE

Certified that this Report titled **". Autonomous Robotic Warehouse Inventory & Micro-Fulfillment System"** is the bonafide work of

Shanmugapriyan D	**722825243197**

who carried out the project work under my supervision.

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SIGNATURE	**SIGNATURE**
Dr.G.Sathish Kumar **HEAD OF THE DEPARTMENT** Artificial Intelligence and Data Science, Sri Eshwar College of Engineering, Coimbatore – 641 202.	Dr.C.Rajeshkumar **SUPERVISOR** Assistant Professor Artificial Intelligence and Data science, Sri Eshwar College of Engineering, Coimbatore – 641 202.

Submitted for the **Autonomous Semester End Mini Project Viva-Voce** held on ……..

INTERNAL EXAMINER	**EXTERNAL EXAMINER**

DECLARATION

Shanmugapriyan D [722825243197]

To declare that the project entitled **"E-Governance Grievance Redressal & Escalation Workflow System"** submitted in partial fulfilment to the University as the project work of Bachelor of Engineering (Artificial Intelligence and Data Science) Degree, is a record of original work done by us under the supervision and guidance of **Dr.C.Rajeshkumar,** Assistant Professor, Department of Artificial Intelligence and Data Science, Sri Eshwar College of Engineering, Coimbatore.

Place: Coimbatore
Date:

[Shanmugapriyan D]

Project Guided by,

[Dr.C.Rajeshkumar, AP/AD]

ACKNOWLEDGEMENT

The success of a work depends on a team and cooperation. I take this opportunity to express my gratitude and thanks to everyone who helped me in my project. I would like to thank the management for the constant support provided by them to complete this project.

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We are indebted to our **Director Mr. R. Rajaram,** for motivating and providing us with all facilities.

I wish to express my sincere regards and deep sense of gratitude to **Dr. Sudha Mohanram, M.E, Ph.D. Principal**, for the excellent facilities and encouragement provided during the course of the study and project.

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I solemnly express our thanks to all the teaching and nonteaching staff of the Artificial Intelligence and Data Science Engineering Department, family and friends for their valuable support which inspired us to work on this project.

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ABSTRACT

This paper presents the design and implementation of an **E-Governance Grievance Redressal & Escalation Workflow System** using **Spring Boot**, a powerful Java framework for developing enterprise web applications. The system provides a centralized platform for registering, tracking, managing, and resolving citizen grievances through a structured digital workflow. It facilitates **CRUD (Create, Read, Update, Delete)** operations for citizen records, grievance management, and administrative data, allowing government departments to efficiently monitor complaints and their current status. A key feature of the system is its **automatic 72-hour escalation mechanism**, which identifies grievances that remain unresolved beyond the defined time limit and automatically changes their status to **ESCALATED**, reducing the need for manual monitoring and improving accountability. The system uses **Spring Data JPA and MySQL** for reliable data persistence and management. The user interface is developed using **HTML, CSS, JavaScript, and Thymeleaf**, providing dashboards, grievance management pages, citizen management, accounting, and reporting features. The proposed system provides a scalable foundation for improving transparency, timely grievance resolution, and efficient digital public-service administration.

1.1 OBJECTIVE

The **E-Governance Grievance Redressal & Escalation Workflow System** aims to develop a robust web-based application for efficiently managing citizen grievances through a centralized digital platform. It focuses on accurate registration, tracking, status management, and timely resolution of grievances while providing administrators with a structured workflow for monitoring pending complaints. Users can manage citizen records, grievances, and administrative information through seamless **CRUD (Create, Read, Update, Delete)** operations, enabling government departments to maintain organized and accessible grievance records.

Built with **HTML, CSS, JavaScript, Thymeleaf, and Spring Boot**, the application provides an intuitive interface for registering grievances, updating their status, managing citizen information, and monitoring grievance progress. A key feature is the **automatic 72-hour escalation mechanism**, which identifies grievances that remain in the OPEN state beyond the specified time period and automatically changes their status to **ESCALATED**. This improves operational visibility, reduces manual follow-up, and helps ensure that pending grievances receive appropriate attention. Data persistence and integrity are maintained using **Spring Data JPA with a MySQL database**.

Ultimately, this project provides government administrators and grievance-handling personnel with a unified platform for coordinating citizen complaints, monitoring resolution progress, identifying overdue grievances, and generating basic reports. The system establishes a foundation for improving **transparency, accountability, timely grievance resolution, and efficient digital governance services**.

2.1 EXISTING SCENARIO

In the current public-service environment, many grievance-handling processes rely on manual registration, paper-based records, telephone communication, emails, or disconnected digital platforms. Although some government organizations provide online grievance portals, the processes may still involve manual monitoring, delayed follow-up, and limited integration between grievance registration, status tracking, and escalation. Administrators may need to manually identify pending complaints and contact the responsible department when grievances remain unresolved for an extended period.

For citizens and smaller administrative units that require a simple, customizable, and centralized grievance management platform, these conventional approaches can be difficult to manage efficiently. Manual spreadsheets and isolated records can make it challenging to maintain accurate grievance histories, monitor complaint status, and identify overdue cases. This gap is where a **Java Spring Boot-based E-Governance Grievance Redressal & Escalation Workflow System** becomes relevant — combining centralized citizen and grievance management with automated time-based escalation. The system provides administrators with greater control over grievance records, status monitoring, reporting, and follow-up while reducing dependence on manual tracking and improving transparency in the grievance redressal process.

2.2 PROBLEM STATEMENT

Government departments and public-service organizations often handle citizen grievances through disconnected channels such as paper forms, emails, telephone calls, spreadsheets, and separate online portals. These approaches can make it difficult to maintain a centralized record of complaints, monitor grievance status, identify pending cases, and ensure that unresolved grievances receive timely attention. Administrators and grievance-handling officers may have to manually track complaints and follow up with responsible departments, resulting in delays, inconsistent record keeping, and limited visibility into the overall grievance redressal process.

This project addresses these challenges by consolidating **citizen management, grievance registration, status tracking, automatic escalation, administrative data management, and reporting** within a single Spring Boot application. Administrators can register and manage citizen records, create and update grievances, monitor their current status, and review grievance information through a centralized dashboard. A scheduled escalation mechanism automatically identifies grievances that remain in the **OPEN** state for more than **72 hours** and changes their status to **ESCALATED**, reducing the need for continuous manual monitoring.

This unified approach provides a common source of information for grievance-handling personnel and administrators, minimizes data silos, and improves visibility into pending and escalated complaints. By combining **Spring Boot, Spring Data JPA, MySQL, Thymeleaf, HTML, CSS, and JavaScript**, the system establishes a scalable foundation for more transparent, organized, and timely digital grievance redressal.

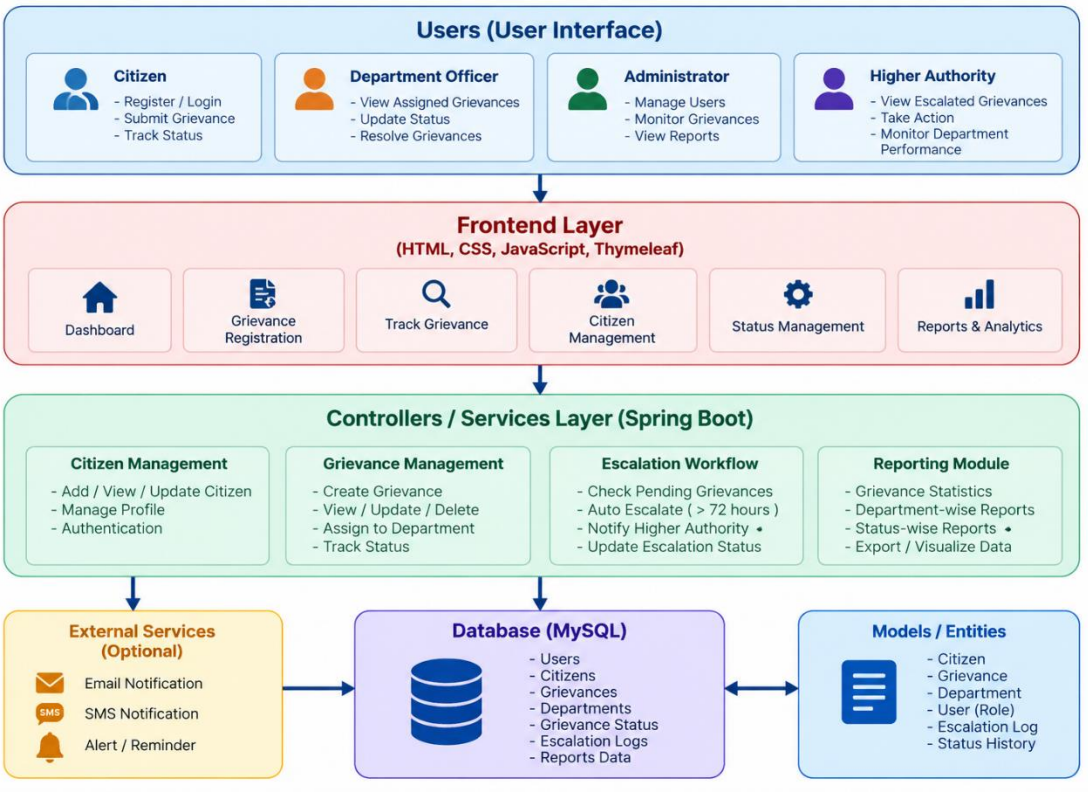
PROPOSED SOLUTION

3.1 OVERVIEW

The **Spring Boot-based E-Governance Grievance Redressal & Escalation Workflow System** provides a centralized platform for managing citizen information and grievance redressal activities. It uses REST controllers to expose grievance management endpoints such as /api/grievances for creating, retrieving, updating, and deleting grievance records, backed by **Spring Data JPA repositories and a MySQL database**. Administrators can register citizens through the **Citizen Management** module, maintain citizen details, and manage grievances by recording their title, description, status, citizen information, and department details. The system supports a structured grievance lifecycle in which newly submitted complaints are assigned an **OPEN** status and can subsequently be updated based on their resolution progress. REST APIs provide the communication layer between the frontend and backend, enabling efficient data management and retrieval.

The frontend, developed using **HTML, CSS, JavaScript, and Thymeleaf**, provides dashboard, grievance management, citizen management, accounting, and reporting interfaces. The dashboard displays recent grievance information, while the reports module provides status-based grievance statistics and visualizations. Overall, the system combines centralized data management, REST-based communication, automated escalation, and reporting to support a more organized and transparent digital grievance redressal process.

3.2 FLOW CHART



4.1 CODE

CONTROLLER — GrievanceApiController.java

```
package com.example.E_Governance_Grievance_Redressal.controller.api;

import com.example.E_Governance_Grievance_Redressal.model.Grievance;
import com.example.E_Governance_Grievance_Redressal.service.GrievanceService;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;

import java.util.List;

@RestController
@RequestMapping("/api/grievances")
public class GrievanceApiController {

@Autowired
private GrievanceService grievanceService;

@PostMapping
public Grievance lodgeGrievance(@RequestBody Grievance grievance) {
return grievanceService.lodgeGrievance(grievance);
}

@GetMapping
public List<Grievance> getAllGrievances() {
return grievanceService.getAllGrievances();
}

@PutMapping("/{id}")
public Grievance updateGrievance(
@PathVariable Long id,
@RequestBody Grievance grievance) {
```

```
    return grievanceService.updateGrievance(id, grievance);
    }

    @DeleteMapping("/{id}")
    public void deleteGrievance(@PathVariable Long id) {
    grievanceService.deleteGrievance(id);
    }
}
```

APPLICATION CONFIG — application.properties

```
spring.application.name=E-Governance_Grievance_Redressal

# Database Configuration
spring.datasource.url=jdbc:mysql://localhost:3306/egov_db?createDatabaseIfNotExist=
true&useSSL=false&allowPublicKeyRetrieval=true
spring.datasource.username=root
spring.datasource.password=YOUR_PASSWORD
spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver

# JPA / Hibernate Configuration
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQLDialect
spring.jpa.properties.hibernate.format_sql=true

# Thymeleaf Configuration
spring.thymeleaf.cache=false

# Task Scheduling for Auto-escalation
spring.task.scheduling.pool.size=5
```

ENTITY — Grievance.java

```
package com.example.E_Governance_Grievance_Redressal.model;

import jakarta.persistence.*;
import lombok.Data;

import java.time.LocalDateTime;
@Entity
```

```
@Data

public class Grievance {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String title;

    private String description;

    // OPEN, RESOLVED, ESCALATED
    private String status;

    @ManyToOne
    private User citizen;

    @ManyToOne
    private AnalyticAccount department;

    private LocalDateTime createdAt = LocalDateTime.now();

    private LocalDateTime escalatedAt;
}
```

INDEX / GRIEVANCES HTML
grievances.html

```
        <!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org"
      xmlns:layout="http://www.ultraq.net.nz/thymeleaf/layout"
      layout:decorate="~{layout}">

<body>

<div layout:fragment="content">

    <h1 class="page-title animated-fade">
```

```
E-Governance Grievance Management
  </h1>

  <!-- Grievance Form -->
  <div class="glass-panel animated-fade delay-1">

    <h3>Submit / Update Grievance</h3>

    <form id="grievanceForm">

      <input
        type="hidden"
        id="grievanceId"
      />

      <input
        type="text"
        id="title"
        placeholder="Grievance Title"
        required
      />

      <textarea
        id="description"
        placeholder="Enter grievance description"
        required>
      </textarea>

      <div>

        <button type="submit">
          Save Grievance
        </button>

        <button
          type="button"
          onclick="resetForm()">
          Clear
        </button>
```

```
        </div>

    </form>

</div>


<!-- Grievance Table -->
<div class="glass-panel">

    <h3>All Grievances</h3>

    <table
        class="styled-table"
        id="grievanceTable">

        <thead>

            <tr>
                <th>ID</th>
                <th>Title</th>
                <th>Description</th>
                <th>Status</th>
                <th>Created At</th>
                <th>Actions</th>
            </tr>

        </thead>

        <tbody id="grievanceTableBody">

            <tr>
                <td colspan="6">
                    Loading...
                </td>
            </tr>

        </tbody>
```

```
        </table>

    </div>

</div>

</body>
</html>
```

JAVASCRIPT — grievances.js

```
const API = "/api/grievances";
let grievances = [];

document.addEventListener("DOMContentLoaded", load);

function load() {
    fetch(API)
        .then(res => res.json())
        .then(data => {
            grievances = data;
            display(data);
        });
}

function display(data) {
    const tbody = document.querySelector("#grievanceTableBody");
    tbody.innerHTML = "";

    data.forEach(g => {
        tbody.innerHTML += `
        <tr>
            <td>${g.id}</td>
            <td>${g.title}</td>
            <td>${g.description}</td>
            <td>${g.status}</td>
            <td>
                <button onclick="edit(${g.id})">Edit</button>
```

```
            <button onclick="remove(${g.id})">Delete</button>
        </td>

    </tr>`;
    });
}

document.querySelector("#grievanceForm")
    .addEventListener("submit", e => {
        e.preventDefault();

        const id = document.querySelector("#grievanceId").value;

        const data = {
            title: document.querySelector("#title").value,
            description: document.querySelector("#description").value
        };

        fetch(id ? `${API}/${id}` : API, {
            method: id ? "PUT" : "POST",
            headers: {"Content-Type": "application/json"},
            body: JSON.stringify(data)
        }).then(() => {
            reset();
            load();
        });
    });

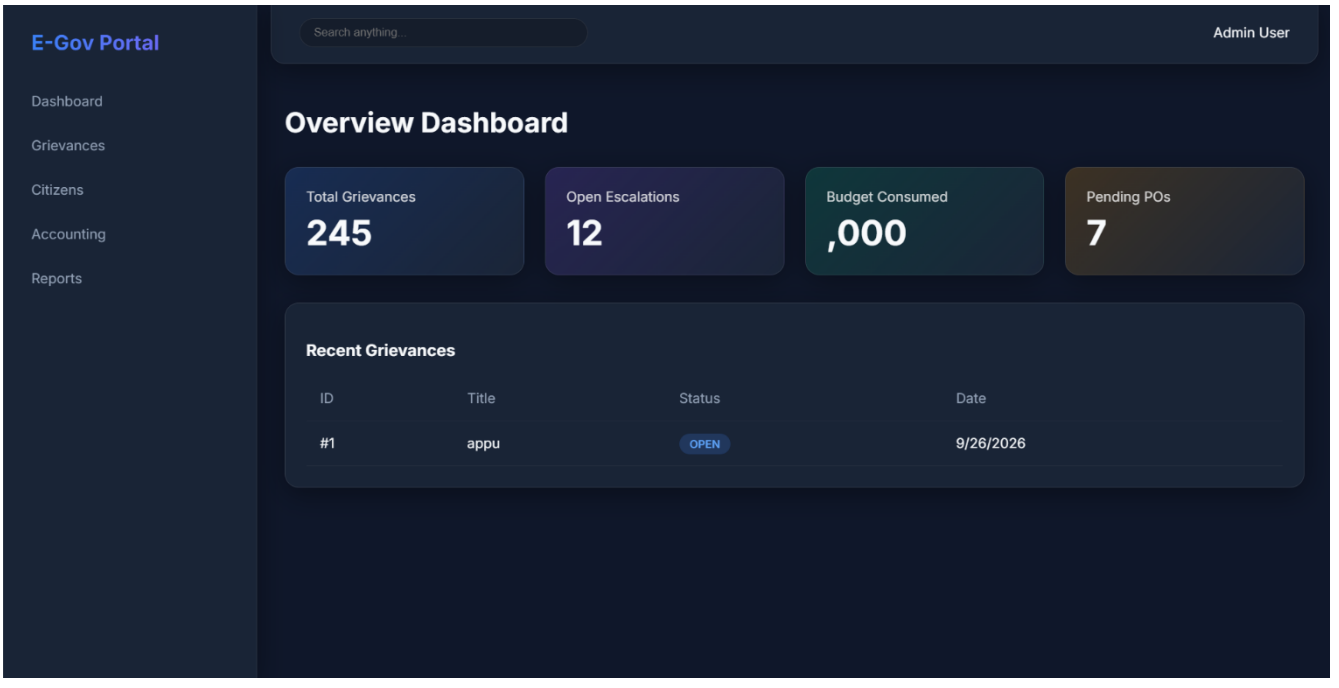
function edit(id) {
    const g = grievances.find(x => x.id === id);

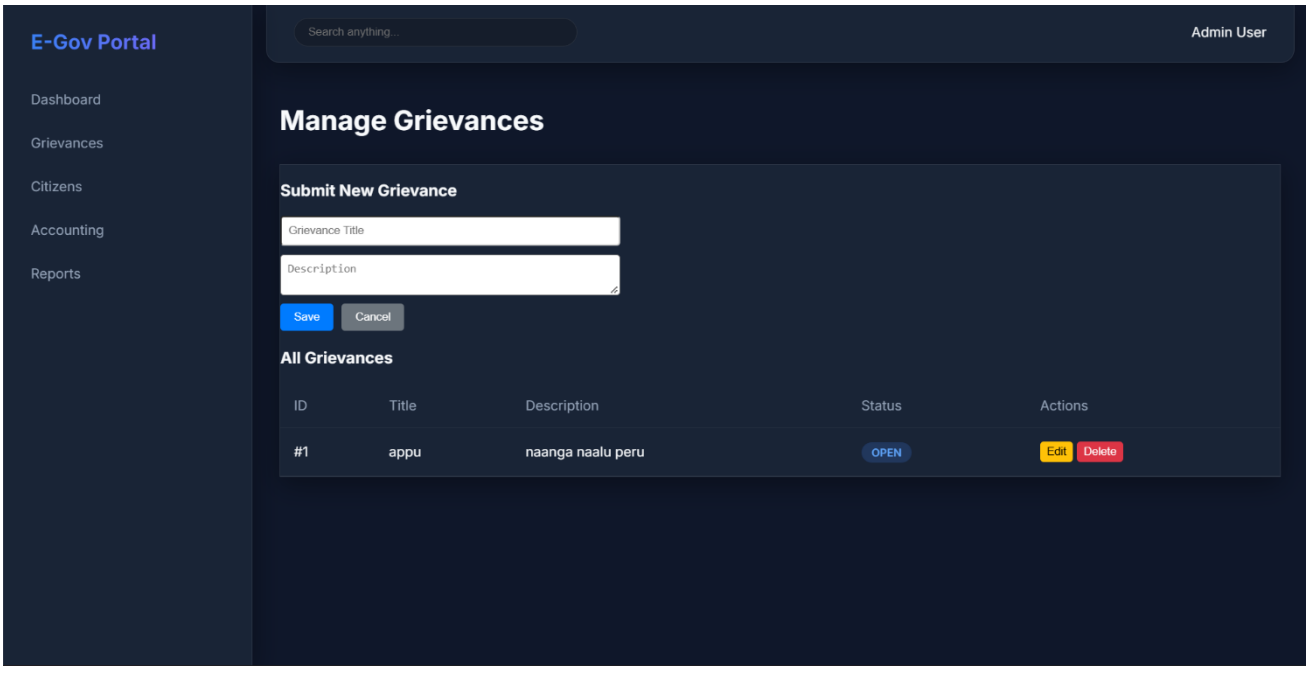
    document.querySelector("#grievanceId").value = g.id;
    document.querySelector("#title").value = g.title;
    document.querySelector("#description").value = g.description;
}
```

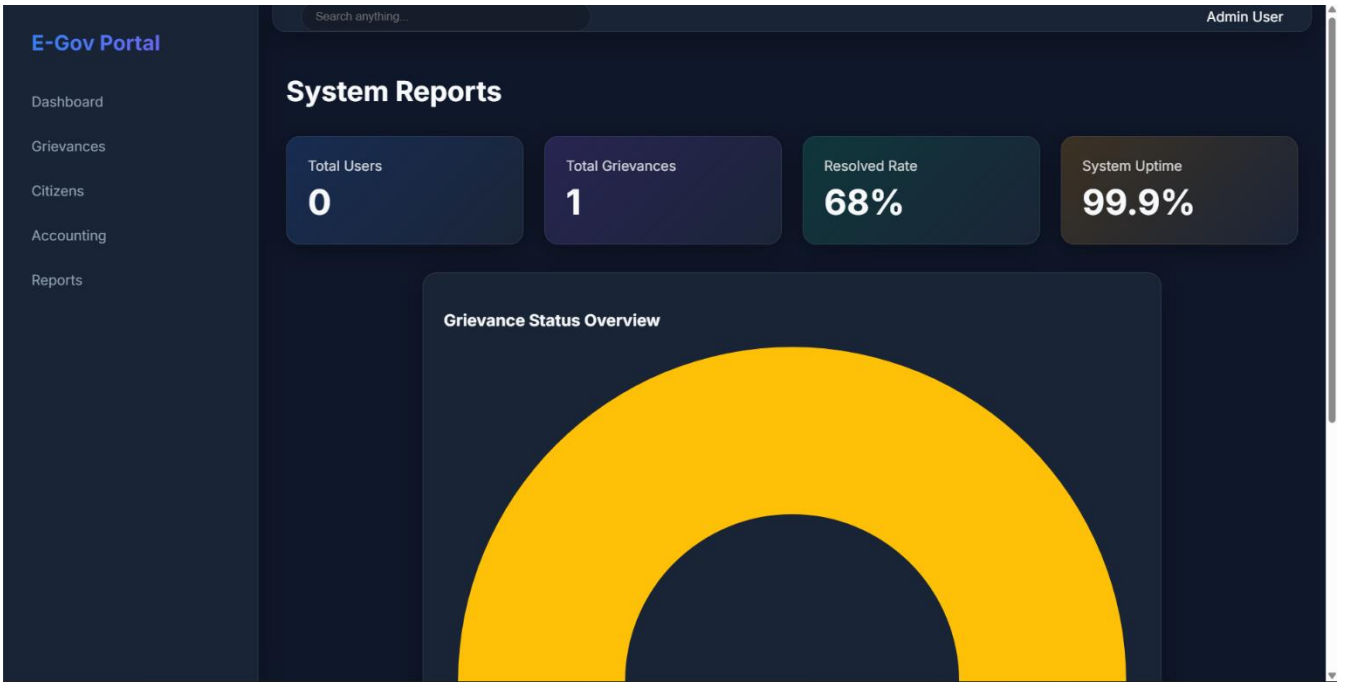
```
function remove(id) {
    fetch(`${API}/${id}`, {method: "DELETE"})
        .then(() => load());
}

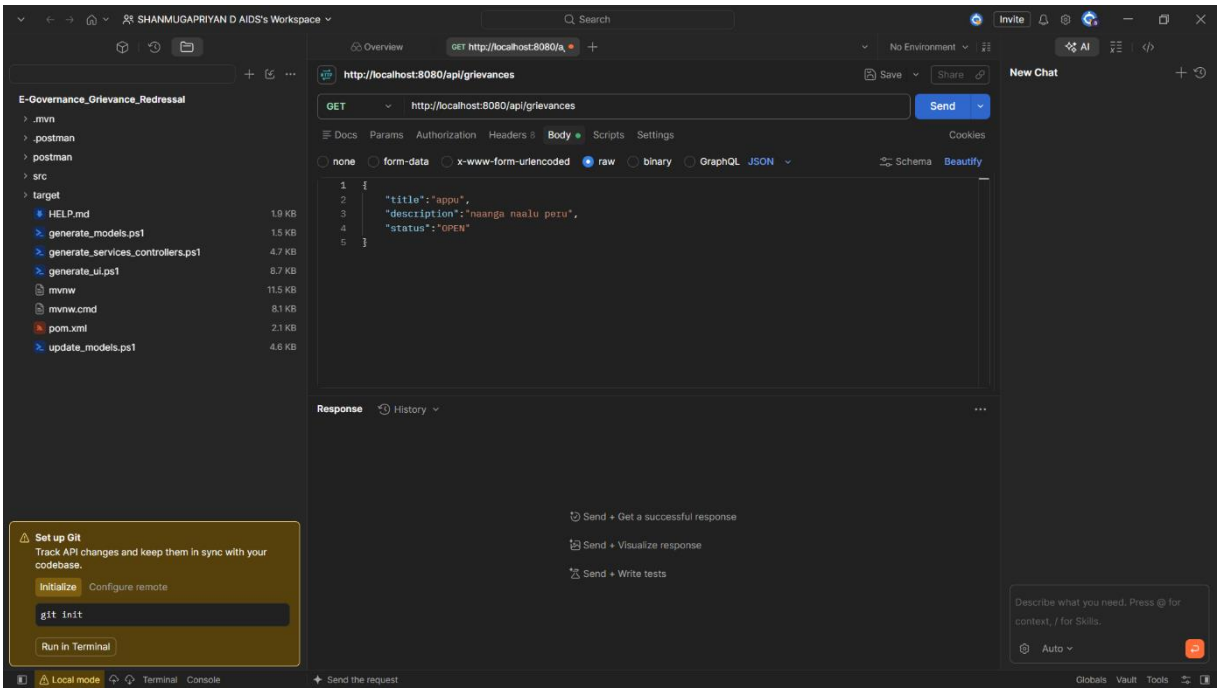
function reset() {
    document.querySelector("#grievanceId").value = "";
    document.querySelector("#title").value = "";
    document.querySelector("#description").value = "";
}
```

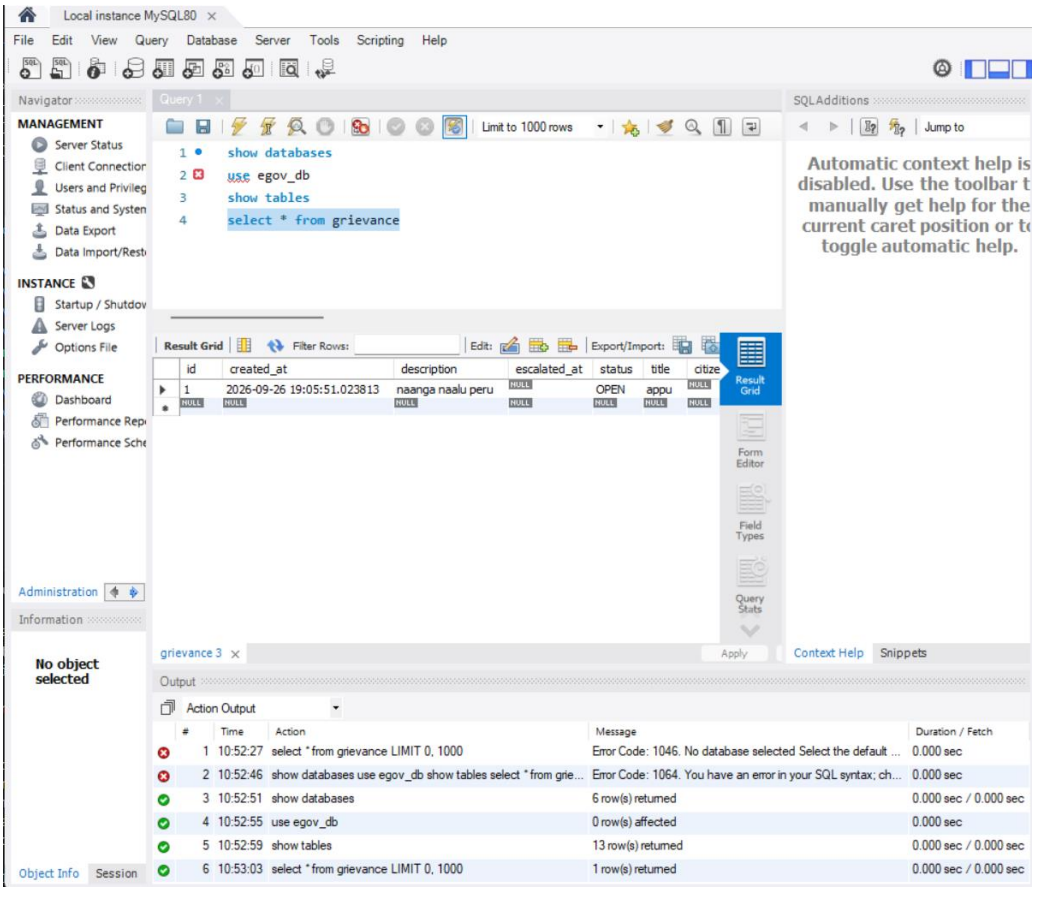
4.2 OUTPUT




E-Gov Portal
Dashboard
Grievances
Citizens
Accounting
Reports
Search anything...
Admin User
Manage Grievances
Submit New Grievance
Grievance Title
Description
Save
Cancel
All Grievances
ID
Title
Description
Status
Actions
#1
appu
naanga naalu peru
OPEN
Edit
Delete


E-Gov Portal
Dashboard
Grievances
Citizens
Accounting
Reports
Search anything...
Admin User
System Reports
Total Users
0
Total Grievances
1
Resolved Rate
68%
System Uptime
99.9%
Grievance Status Overview


SHANMUGAPRIYAN D AIDS's Workspace
Search
Invite
E-Governance_Grievance_Redressal
.mvn
.postman
postman
src
target
HELP.md 1.9 KB
generate_models.ps1 1.5 KB
generate_services_controllers.ps1 4.7 KB
generate_ui.ps1 8.7 KB
mvnw 11.5 KB
mvnw.cmd 8.1 KB
pom.xml 2.1 KB
update_models.ps1 4.6 KB
http://localhost:8080/api/grievances
GET
http://localhost:8080/api/grievances
Send
Docs Params Authorization Headers 8 Body Scripts Settings
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
{
"title":"appu",
"description":"naanga naalu peru",
"status":"OPEN"
}
Response History
Send + Get a successful response
Send + Visualize response
Send + Write tests
New Chat
Describe what you need. Press @ for context, / for Skills.
Auto
Set up Git
Track API changes and keep them in sync with your codebase.
Initialize Configure remote
git init
Run in Terminal
Local mode Terminal Console
Send the request
Globals Vault Tools


Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
MANAGEMENT
Server Status
Client Connection
Users and Privileg
Status and Systen
Data Export
Data Import/Rest
INSTANCE
Startup / Shutdov
Server Logs
Options File
PERFORMANCE
Dashboard
Performance Rep
Performance Sche
Administration
Information
No object selected
Object Info Session
Query 1
Limit to 1000 rows
1 show databases
2 use egov_db
3 show tables
4 select * from grievance
Result Grid Filter Rows: Edit: Export/Import:
id created_at description escalated_at status title citize
1 2026-09-26 19:05:51.023813 naanga naalu peru NULL OPEN appu NULL
NULL NULL NULL NULL NULL NULL NULL
Result Grid
Form Editor
Field Types
Query Stats
grievance 3
Apply
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar t manually get help for the current caret position or t toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
1 10:52:27 select * from grievance LIMIT 0, 1000 Error Code: 1046. No database selected Select the default ... 0.000 sec
2 10:52:46 show databases use egov_db show tables select * from grie... Error Code: 1064. You have an error in your SQL syntax; ch... 0.000 sec
3 10:52:51 show databases 6 row(s) returned 0.000 sec / 0.000 sec
4 10:52:55 use egov_db 0 row(s) affected 0.000 sec
5 10:52:59 show tables 13 row(s) returned 0.000 sec / 0.000 sec
6 10:53:03 select * from grievance LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec

CONCLUSION

The E-Governance Grievance Redressal & Escalation Workflow System successfully demonstrates how a Java Spring Boot backend, combined with Spring Data JPA and a MySQL database, can provide a centralized platform for managing citizen grievances and administrative records. The project addresses the major challenge of disconnected and manually monitored grievance processes by bringing grievance registration, citizen management, status tracking, and automated escalation into a single web-based application.

The system follows a layered architecture consisting of the presentation layer using **HTML, CSS, JavaScript, and Thymeleaf**, REST controllers, service classes, repositories, JPA entities, and the MySQL database. This separation of responsibilities improves maintainability and provides a foundation for future expansion. The grievance controller, service, repository, and entity components together support CRUD operations for managing grievance records, while the citizen management module provides corresponding operations for maintaining citizen information.

From a functional perspective, the application allows administrators and grievance-handling personnel to register citizens, submit grievances, update grievance information, monitor complaint status, and delete records when required. A key feature of the system is the **automatic 72-hour escalation mechanism**. Grievances that remain in the **OPEN** state beyond the defined time limit are automatically identified by the scheduled service and changed to **ESCALATED**, with the escalation time recorded in the database. This reduces the need for continuous manual monitoring and improves visibility of pending complaints.

The system also provides dashboard and reporting interfaces for viewing grievance information and status-based statistics. The REST API architecture allows the frontend to communicate independently with the backend, demonstrating a decoupled application structure that can be extended to other clients in the future. MySQL provides persistent storage, while Spring Data JPA simplifies database interaction and maintains structured relationships between citizens, grievances, and administrative entities.

Overall, this project fulfills its objective of developing a **centralized, customizable, and cost-effective digital grievance management platform**. By combining citizen management, grievance tracking, automated escalation, and reporting, the system provides a strong foundation for improving **transparency, accountability, organized record management, and timely grievance redressal** in e-governance applications.

FUTURE SCOPE

While the current implementation provides the core grievance registration, management, tracking, and automatic escalation workflow, several enhancements can further improve the system. **Role-based authentication and authorization** can be introduced for citizens, department officers, administrators, and higher-level authorities. SMS, e-mail, and push notifications can be integrated to inform citizens and officials about grievance registration, status changes, and escalations.

The system can also be enhanced with **department-wise grievance assignment and multi-level escalation**, such as Officer → Department Head → Higher Authority. A unique grievance reference number, document/image attachments, grievance priority, location/GIS support, and complete audit history can improve usability and accountability. Advanced reporting can provide statistics such as average resolution time, department-wise performance, pending grievance trends, and SLA compliance.

Finally, integration with government service portals, mobile applications, secure cloud deployment, automated testing, database backup, and real-time analytics can transform the current application into a more comprehensive **production-ready e-governance grievance redressal platform**.

REFERENCE

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Sri Eshwar College of Engineering
Coimbatore | Tamilnadu
An Autonomous Institution
Affiliated to Anna University, Chennai

PROJECT TITLE	**E-Governance Grievance Redressal & Escalation Workflow System**
PROGRAM	**B.TECH ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**
NAME OF THE SUPERVISOR	**Dr.C.Rajeshkumar**
NAME OF THE SDG GOALS MAPPED	**Peace, Justice and Strong Institutions; Industry, Innovation and Infrastructure; Sustainable Cities and Communities**
MENTION THE SDG GOALS NUMBER	**SDG 16, SDG 9, SDG 11**
NAME OF THE TRL LEVEL	**Technology Demonstration**
MENTION THE TRL LEVEL	**TRL 6**

POs & PSOs Mapping (Put a tick mark in the mapped PO's & PSO's):

Program Outcomes											Program Specific Outcomes		
PO1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3

Signature of the Supervisor
(Dr.C.Rajeshkumar)

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Coimbatore | Tamilnadu
An Autonomous Institution
Affiliated to Anna University, Chennai

VENUE AND EXPENDITURE STATEMENT FOR THE PROJECT WORK

Laboratory details where the project is carried out	**DVA LAB**
Software / Hardware details	• **Operating System: Windows 10/11 (64-bit)** • **IDE: IntelliJ IDEA** • **Programming Language: Java (JDK 17)** • **Framework: Spring Boot 3.x** • **Build Tool: Maven** • **ORM: Spring Data JPA / Hibernate** • **Database: MySQL 8.x** • **Frontend: HTML5, CSS3, JavaScript** • **API Testing: request.http / Postman** • **Version Control: Git & GitHub** • **Browser: Google Chrome**

Details of the Component and Expenditure

S. No	Name of the Component	Qty	Price / Unit in (Rs.)	Amount (Rs.)

(*Include any other charges, which includes fabrication cost and others)

Signature of the student
(Shanmugapriyan D)

Signature of the Supervisor
(Dr.C.Rajeshkumar)